158.888 Information Technology Professional Project

Final Project Report on

Real-Time Parking Occupancy Monitoring and Area-Level Prediction Using Spatio-Temporal Analytics

Presented by

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Abstract

Urban parking congestion is a major challenge in modern cities, contributing to traffic delays, fuel consumption, pollution, and driver frustration. Smart parking systems have become an important area of smart city development, aiming to improve parking management through real-time data and predictive analytics. This project presents the development of a machine learning-based smart parking occupancy prediction and visualization system that combines historical parking data, weather information, and live parking feeds to forecast short-term parking availability.

The project was completed in two phases. The first phase focused on exploratory analysis using public parking datasets from Geelong, Melbourne, and Brisbane in Australia. During this stage, different pre-processing methods, feature engineering techniques, and machine learning models were evaluated to better understand parking occupancy behaviour across different cities. Temporal features such as hour, weekday, weekend, and seasonal patterns were analysed together with weather-related variables including temperature and rainfall. Cross-city prediction and calibration experiments were also conducted to examine whether parking patterns learned from one city could be transferred to another.

The second phase focused on developing a localized smart parking prediction system for Auckland, New Zealand. Historical parking data and real-time occupancy feeds from Auckland Transport were integrated into a complete forecasting pipeline. The system included data pre-processing, feature engineering, occupancy prediction, and an interactive dashboard.

Several machine learning models, including Linear Regression, Random Forest, and LightGBM, were evaluated during implementation. Time-based validation was used to simulate realistic forecasting conditions and reduce temporal data leakage. The final LightGBM framework achieved MAE values ranging from 0.1214 to 0.1298 across validation experiments.

The completed system demonstrates how machine learning and urban data analytics can support smarter parking management and improve urban mobility. The project also highlights practical challenges associated with heterogeneous datasets, limited live sensor coverage, and operational real-time forecasting.

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1. Introduction

1.1 Background

Rapid urbanization and the increasing number of vehicles have created significant parking and transportation challenges in modern cities. In busy urban areas, drivers often spend a large amount of time searching for available parking spaces, particularly in commercial and central business districts. This contributes to traffic congestion, fuel waste, environmental pollution, and driver frustration while also reducing the efficiency of urban transportation systems.

As smart city technologies continue to develop, smart parking systems have become an important part of intelligent transportation infrastructure. Traditional parking management methods mainly rely on static parking signs, manual monitoring, or simple occupancy counters. Although these methods provide basic parking information, they cannot predict future parking availability or support proactive parking management.

Recent advances in machine learning and urban data analytics have created opportunities to develop predictive parking systems using historical parking behaviour, weather conditions, temporal patterns, and real-time occupancy information. Parking occupancy often follows recurring patterns influenced by factors such as time of day, weekdays, weekends, weather conditions, and surrounding urban activity. By learning these patterns from historical data, machine learning models can estimate near-future parking occupancy levels and support smarter parking management decisions.

This project explores how machine learning and real-time urban data can be combined to develop a parking occupancy prediction and visualization framework. The system integrates data pre-processing, feature engineering, machine learning modelling, live occupancy integration, and dashboard visualization while addressing practical challenges such as heterogeneous datasets and limited real-time sensor coverage.

1.2 Problem Statement

Parking congestion remains a major problem in urban transportation systems, increasing traffic congestion, fuel consumption, carbon emissions, and driver frustration. Although many cities collect parking occupancy data, developing accurate forecasting systems remains challenging due to inconsistent datasets, changing urban activity patterns, weather influences, and limited real-time sensor coverage. Existing parking systems mainly provide static occupancy information and often lack predictive capability. Another challenge is combining historical parking behaviour with live occupancy feeds in a practical and scalable way. This project addresses these issues by developing a machine learning-based smart parking prediction framework that integrates historical parking data, temporal features, weather information, and real-time occupancy feeds to forecast near-future parking conditions.

1.3 Aim and Objectives

The main aim of this project is to design and develop a smart parking occupancy prediction and visualization system using machine learning and real-time urban data.

The objectives of the project are:

- to analyse historical parking occupancy behaviour,
- to identify useful predictive features,
- to forecast near-future parking occupancy levels,
- to integrate live parking occupancy feeds,
- to evaluate multiple machine learning models,
- and to visualize parking conditions through an interactive dashboard.

The project also explores the practical challenges involved in developing operational smart parking systems using heterogeneous urban datasets collected from different cities.

1.4 Research Questions

This project is guided by the following research questions:

1. Can machine learning models accurately predict near-future parking occupancy using historical parking data?
2. How do temporal and weather-related features influence parking occupancy prediction performance?
3. Which machine learning model provides the most suitable forecasting performance for smart parking occupancy prediction?

1.5 Scope of the Project

This project focuses on near-future parking occupancy forecasting within a smart city environment. The work includes both exploratory research and operational system implementation.

The exploratory phase used parking datasets collected from Australian cities including Geelong, Melbourne, and Brisbane. This stage focused on occupancy behaviour analysis, feature engineering, and evaluation of machine learning approaches across multiple urban datasets.

The final implementation phase focused on Auckland, New Zealand. Historical Auckland parking datasets were combined with live parking occupancy information obtained from Auckland Transport to develop a real-time parking prediction and visualization system. The project mainly focuses on one-hour-ahead occupancy prediction and selected live parking locations because of real-time data availability constraints.

Although the system demonstrates real-time forecasting capability, it is designed as a prototype smart parking framework rather than a fully deployed production-level system.

2. Literature Review

2.1 Smart Parking Systems and Urban Mobility

Rapid urbanisation and increasing vehicle ownership have placed significant pressure on urban transportation systems worldwide. One of the most persistent challenges associated with urban mobility is parking management. Drivers searching for parking spaces contribute to unnecessary traffic circulation, increased fuel consumption, environmental pollution, and reduced transportation efficiency. As cities continue to grow, parking management has become an important component of broader smart city initiatives aimed at improving mobility, sustainability, and the quality of urban services (Batty et al., 2012).

Smart parking systems have emerged as a practical response to these challenges. Unlike traditional parking management approaches that rely on static signage and manual monitoring, smart parking systems utilise digital technologies to monitor, analyse, and manage parking resources more effectively. These systems typically integrate sensors, cameras, communication networks, mobile applications, and cloud-based platforms to provide real-time information about parking availability and occupancy conditions (Fahim et al., 2021; Knights et al., 2024).

The development of smart parking technologies has been closely linked with advances in the Internet of Things (IoT). IoT-enabled parking systems allow parking infrastructure to continuously generate occupancy data that can be transmitted, processed, and analysed in real time. According to Dahiya et al. (2025), the combination of IoT technologies and machine learning techniques has significantly improved the ability of parking systems to optimise resource allocation and reduce parking search times. Similar findings were reported by Knights et al. (2024), who highlighted that intelligent parking platforms are increasingly moving beyond simple occupancy monitoring towards predictive and decision-support capabilities.

Smart parking systems are also recognised as an important component of smart city ecosystems. Smart cities rely on the integration of digital infrastructure, data analytics, and intelligent decision-making to improve urban services and operational efficiency (Batty et al., 2012). Within this context, parking management represents a valuable application area because parking demand directly influences traffic congestion, accessibility, and urban mobility performance. Kitchin (2014) argued that the growing availability of urban data enables cities to move from reactive management approaches towards more proactive and data-driven decision-making.

Despite these advancements, predicting parking occupancy remains a complex challenge. Parking demand is influenced by multiple interacting factors, including temporal patterns, weather conditions, local land use, population density, commercial activity, and transportation behaviour. These factors often vary considerably between locations and over time, making parking occupancy difficult to model using simple rule-based approaches. Consequently, researchers have increasingly explored machine learning and predictive analytics techniques to identify hidden patterns within historical parking datasets and generate accurate occupancy forecasts (Ma et al., 2024).

2.2 Machine Learning for Parking Occupancy Prediction

Machine learning has become one of the most widely adopted approaches for transportation forecasting due to its ability to learn complex relationships from large datasets. In parking management applications, occupancy prediction is commonly formulated as a regression or time-series forecasting problem, where models attempt to estimate future occupancy levels based on historical observations and contextual variables (Ma et al., 2024).

A wide range of machine learning algorithms have been applied to parking occupancy prediction. Traditional approaches such as Linear Regression remain popular because of their simplicity, interpretability, and computational efficiency (James et al., 2021). However, urban parking demand is often influenced by complex and non-linear relationships that may not be fully captured by linear models.

To address these limitations, ensemble learning methods have become increasingly popular. Random Forest, introduced by Breiman (2001), combines multiple decision trees to improve predictive performance and reduce overfitting. More recent approaches such as XGBoost and LightGBM have demonstrated strong performance across structured forecasting datasets because of their ability to model complex feature interactions efficiently (Chen & Guestrin, 2016; Ke et al., 2017). These ensemble methods have been widely adopted in transportation forecasting and intelligent mobility applications due to their balance between predictive performance and operational efficiency (Ma et al., 2024; Zhang et al., 2024).

Previous research suggests that predictive performance depends not only on model selection but also on feature engineering. Temporal variables such as hour of day, day of week, seasonal trends, and weekend indicators consistently appear among the strongest predictors of parking demand (Knights et al., 2025; Ma et al., 2024). These variables capture recurring behavioural patterns that influence parking utilisation throughout the day and across different periods.

Occupancy-history features have also received considerable attention within the literature. Lag variables and rolling occupancy statistics enable forecasting models to incorporate recent parking behaviour when predicting future occupancy levels. Knights et al. (2025) demonstrated that lag-based features significantly improved forecasting performance within IoT-enabled parking environments because recent occupancy conditions often provide strong indications of near-future demand.

Weather conditions represent another important source of predictive information. Studies have reported that temperature, rainfall, and precipitation levels influence travel behaviour and parking demand across urban areas (Ma et al., 2024). Incorporating weather variables alongside temporal and occupancy-history features has therefore become a common practice within smart parking forecasting research.

Recent research has also highlighted challenges associated with transferring forecasting models between different cities and urban environments. Parking demand is influenced by local factors such as land use patterns, transportation infrastructure, population behaviour, and parking regulations. As a result, forecasting models developed in one city may not always generalise effectively when applied elsewhere. Researchers have increasingly recognised the importance of localisation, adaptation, and calibration strategies to improve forecasting performance across different urban contexts (Ma et al., 2024; Knights et al., 2025). This issue is particularly relevant to the present study because one objective of the exploratory phase was to investigate cross-city forecasting behaviour and transferability.

Although machine learning models offer strong predictive capabilities, researchers have identified several practical challenges. Parking datasets often contain missing observations, inconsistent sensor coverage, and heterogeneous data structures. Furthermore, real-world deployment requires models to generate predictions efficiently while maintaining acceptable accuracy levels. These operational considerations have become increasingly important as parking forecasting systems move from research environments into practical smart city applications (Knights et al., 2024).

2.3 Deep Learning, City Analytics, and Real-Time Transportation Intelligence

The rapid growth of urban data has encouraged researchers to investigate more advanced forecasting techniques within intelligent transportation systems. Deep learning approaches have attracted particular attention because of their ability to model complex sequential patterns and high-dimensional datasets. Models such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Transformer-based architectures have been applied to traffic forecasting, mobility prediction, and parking occupancy estimation (Zhang et al., 2024). Among these approaches, LSTM networks have received considerable attention because they can capture temporal dependencies within sequential data (Hochreiter & Schmidhuber, 1997). However, deep learning models generally require larger datasets and greater computational resources than traditional machine learning methods (Zhang et al., 2024).

Consequently, recent research suggests that simpler machine learning models combined with effective feature engineering can often achieve competitive performance, particularly in operational environments where data availability may be limited or fragmented (Knights et al., 2025). This perspective is particularly relevant for practical smart parking systems that must balance forecasting accuracy with deployment complexity, computational efficiency, and maintainability.

The increasing availability of urban data has also strengthened the role of smart city analytics within transportation research. Modern cities generate large volumes of information through parking sensors, connected vehicles, mobile applications, weather services, and transportation infrastructure. These data sources provide valuable opportunities for understanding mobility behaviour and supporting evidence-based urban planning (Kitchin, 2014).

Predictive analytics has become a central component of smart city initiatives because it enables decision-makers to anticipate future demand rather than simply reacting to existing conditions. According to Batty et al. (2012), data-driven urban management represents a key characteristic of smart cities, allowing public authorities to optimise infrastructure utilisation and improve service delivery. Within parking management, predictive analytics can support proactive resource allocation, congestion reduction, and improved parking accessibility.

Geographic Information Systems (GIS) support the visualisation and analysis of spatial parking patterns, helping identify occupancy hotspots and location-specific demand characteristics (Schneble & Kauermann, 2021). Recent studies suggest that combining GIS and machine learning can improve transportation decision support by providing geographically contextualised insights into parking demand.

Another major development within smart transportation research is the emergence of real-time forecasting systems. Traditional forecasting approaches primarily rely on historical datasets and offline analysis. In contrast, real-time systems continuously process incoming data streams and generate dynamic predictions that reflect current operating conditions (Knights et al., 2024).

The expansion of IoT technologies has played a significant role in enabling real-time transportation analytics. Sensors, cameras, and connected devices generate continuous streams of occupancy information that can be integrated into forecasting models. This capability allows parking systems to move beyond descriptive analytics and towards predictive decision-making (Dahiya et al., 2025).

Operational dashboards transform forecasting outputs into accessible visual information through charts, maps, and performance indicators. Business Intelligence platforms such as Power BI allow predictive models and multiple data sources to be integrated into a single decision-support environment.

2.4 Research Gaps and Motivation for the Project

Although significant progress has been made in smart parking research, several important gaps remain within the existing literature.

First, many studies focus primarily on offline forecasting experiments using historical datasets. While these studies provide valuable insights into model performance, relatively limited attention has been given to integrating machine learning forecasting models with live operational parking systems and real-time occupancy feeds (Knights et al., 2024; Dahiya et al., 2025).

Second, the transferability of parking forecasting models between different cities remains insufficiently explored. Parking demand is influenced by local infrastructure, population characteristics, transportation behaviour, and urban design. Consequently, models trained in one city may not generalise effectively when applied to another environment. Existing research has acknowledged this challenge, but practical investigations into cross-city adaptation and calibration remain relatively limited (Ma et al., 2024).

Third, many smart parking studies assume extensive sensor coverage and highly structured datasets. In practice, real-world parking environments often contain incomplete occupancy information, heterogeneous data sources, and varying levels of real-time monitoring capability. These operational constraints create challenges for the deployment of scalable forecasting systems within diverse urban environments.

Finally, while forecasting accuracy remains important, practical deployment considerations such as real-time integration, dashboard visualisation, operational usability, and decision-support functionality are often underrepresented within the literature. There remains a need for studies that bridge the gap between predictive modelling research and deployable smart parking solutions.

Motivated by these gaps, this project develops a machine learning-based smart parking forecasting framework that combines historical parking data, weather information, real-time occupancy feeds, and dashboard visualisation. The study also investigates cross-city forecasting behaviour before developing a localised Auckland implementation.

3. Datasets and Data Sources

This project used multiple datasets collected from Australia and New Zealand to support parking occupancy prediction, exploratory machine learning experiments, and real-time smart parking visualization. The datasets included historical parking occupancy records, real-time parking availability data, weather information, and calendar-related features.

The project used publicly available or institutionally provided parking datasets that did not contain personally identifiable information.

3.1 Australian Parking Datasets

3.1.1 Geelong Parking Dataset

The Geelong parking dataset was collected from the Victorian Government open data platform. The dataset contained real-time parking availability and parking occupancy related information from Geelong, Australia.

This dataset contains approximately 127,000 records and is used for initial exploratory analysis, pre-processing experiments, feature engineering, and early machine learning model development.

Source: Geelong Real-Time Parking Availability Status Dataset

https://discover.data.vic.gov.au/dataset/real-time-parking-availability-status

3.1.2 Melbourne Parking Dataset

The Melbourne dataset contained on-street parking bay sensor information collected from parking sensors across Melbourne, Australia.

This dataset contains approximately 3,300 records and is used for cross-city validation, transferability experiments, and evaluation of generalized parking prediction behaviour.

Source: Melbourne On-Street Parking Bay Sensors Dataset

https://discover.data.vic.gov.au/dataset/on-street-parking-bay-sensors

3.1.3 Brisbane Parking Dataset

The Brisbane dataset contained parking occupancy forecasting related information from Brisbane City Council open data resources.

This dataset contains approximately 1,048,000 records and is used for additional forecasting experiments, simulation testing, and comparative urban occupancy analysis.

Source: Brisbane Parking Occupancy Forecasting Dataset

https://data.brisbane.qld.gov.au/explore/dataset/parking-occupancy-forecasting

3.2 Auckland Parking Datasets

3.2.1 Auckland Historical Parking Dataset

Two historical Auckland parking occupancy datasets were used during the final implementation phase of the project. The first dataset was a smaller public Auckland Transport dataset containing approximately three weeks of historical parking occupancy records from four parking locations with approximately ~2,100 records. The second dataset is comparatively larger, contains ~29,000 records of multiple location's parking history of selected dates from 2015, 2018, 2021 collected from Auckland Transport through email communication for the project.

The processed datasets included occupancy records, parking location identifiers, temporal features, weather information, and engineered occupancy features. These datasets are used for final model training, validation, feature engineering, and Auckland-specific occupancy forecasting.

Source: Auckland Transport historical parking data (public datasets and direct communication through email).

3.2.2 Auckland Real-Time Parking Feed

Real-time parking availability information was collected from Auckland Transport parking availability feeds. The live data provided continuously updated available parking counts for selected Auckland parking facilities.

This feed is used for real-time occupancy monitoring, live occupancy calculation, and near-future parking occupancy predictions.

Source: Auckland Transport live parking availability feed.

https://at.govt.nz/umbraco/surface/parkingavailabilitysurface/ParkingAvailabilityResult

3.3 Weather Data

Weather information was integrated into both the exploratory and final implementation phases of the project. The datasets included temperature, precipitation, rainfall indicators, and rain intensity information.

Used for environmental feature engineering, weather-occupancy analysis, and occupancy forecasting experiments.

Source: Open-Meteo weather API services.

3.4 Calendar and Temporal Data

Calendar and temporal features were generated to capture periodic occupancy behaviour patterns. The project used hour of day, weekday information, weekends, month values, and public holiday indicators.

This data is used for temporal feature engineering, cyclical occupancy modelling, and forecasting patterns analysis.

Source: Public holiday Nager.Date API and generated temporal features from timestamps.

3.5 Data Processing Challenges

Several challenges were encountered while working with the datasets. Since the data originated from multiple cities and different collection systems, pre-processing and standardization were required before machine learning experiments could be conducted.

The main challenges included inconsistent timestamp formats, missing occupancy records, differing parking structures across cities, limited real-time coverage, and inconsistent geographic metadata. These challenges influenced both the exploratory research phase and the design of the final Auckland smart parking forecasting system.

4. Exploratory Cross-City Experiments

4.1 Overview of the Exploratory Phase

Before developing the final Auckland smart parking prediction system, exploratory experiments were conducted using parking datasets from Geelong, Melbourne and Brisbane. The purpose was to evaluate whether machine learning models could learn parking occupancy behaviour and generalize across different cities, particularly because historical Auckland parking data was limited. Geelong and Melbourne were used for cross-city validation experiments, while Brisbane data provided additional exploratory insights into parking occupancy behaviour and feature engineering.

The exploratory phase focused on parking data pre-processing, feature engineering, weather integration, machine learning experimentation, cross-city validation, and calibration-based forecasting improvement.

4.2 Data Processing and Occupancy Preparation

The Australian parking datasets contained different formats and structures, requiring extensive pre-processing before machine learning analysis could be performed.

The workflow included:

- cleaning and standardizing raw parking data,
- converting timestamps into date time formats,
- aggregating parking behaviour into hourly intervals,
- handling missing values,
- and generating normalized occupancy rates.

Initial exploration showed clear temporal occupancy patterns across weekdays, weekends, and peak-hour periods, highlighting the importance of temporal feature engineering.

4.3 Temporal and Weather Feature Engineering

Temporal, environmental, and occupancy-history features were engineered to improve forecasting performance. These included:

- hour and weekday features,
- weather-related variables,
- cyclical time encoding,
- lag features,
- and rolling occupancy statistics.

The experiments demonstrated that temporal behaviour and recent occupancy history were among the strongest predictors of parking demand.

4.4 Initial Machine Learning Experiments

The first forecasting experiments were conducted using the Geelong dataset. Parking occupancy prediction was treated as a regression problem using engineered temporal and contextual features. An XGBoost regressor was used as an initial forecasting model because of its strong performance on structured tabular datasets and achieved **MAE: 0.2168** and **RMSE: 0.2968.** These results confirmed that machine learning models could successfully capture meaningful parking occupancy behaviour from historical urban parking data.

4.5 Cross-City Validation Experiments

Cross-city validation experiments were conducted to evaluate whether parking patterns learned from one city could generalize to another environment. The Geelong-trained model was tested using Melbourne parking data.

Geelong to Melbourne Validation Results

Metric Values: **MAE 0.4671** and **RMSE 0.5268**

The forecasting performance declined significantly, showing that parking behaviour differs across cities and that direct model transferability is limited without adaptation.

4.6 Auckland Transferability and Calibration Experiments

Additional experiments were conducted using Auckland parking data to investigate localized adaptation.

Geelong to Auckland Baseline Results

Metric Values: **MAE 0.3346** and **RMSE 0.3972**

Residual-based calibration techniques were then applied using Auckland-specific samples. The calibration approach adjusted model predictions using error patterns observed from a limited subset of Auckland data rather than retraining the forecasting model from scratch.

Geelong to Auckland Calibrated Results

Metric Values: **MAE 0.0402** and **RMSE 0.0566**

The calibration process substantially improved forecasting accuracy, demonstrating that limited local information can significantly improve forecasting performance when transferring models between different urban environments.

These findings motivated a shift toward localized forecasting and contributed to the development of the final Auckland prediction framework.

4.7 Evolution of the Forecasting Framework

As experimentation progressed, the forecasting framework evolved into a more structured time-series prediction system.

Key improvements included:

- next-hour target generation
- lag feature engineering
- rolling occupancy statistics
- interaction features
- chronological time-based validation

Additional benchmarking experiments using Random Forest, Ridge Regression, and LightGBM were also performed during this phase.

4.8 Key Findings from the Exploratory Phase

The exploratory phase produced several important findings that influenced the final Auckland implementation:

- temporal occupancy behaviour was the strongest forecasting factor
- weather-related variables improved prediction performance
- direct cross-city transferability was limited
- localized calibration significantly improved forecasting accuracy

These findings directly shaped the final Auckland smart parking forecasting framework.

5. Auckland Smart Parking System Architecture and Operational Design

5.1 Overview of the Auckland Implementation Phase

Following the exploratory cross-city experiments using Australian parking datasets, the project evolved into a more operational smart parking forecasting system focused on Auckland, New Zealand. This transition was influenced by two main factors. First, the exploratory experiments showed that parking occupancy behaviour varied significantly between cities, limiting the effectiveness of directly transferring models across urban environments. Second, access to Auckland Transport historical datasets and live parking feeds became available during the later stages of the project.

The Auckland phase focused on developing a localized smart parking system capable of processing historical occupancy behaviour, integrating temporal and weather information, generating near-future occupancy forecasts, incorporating real-time parking feeds, and visualizing parking conditions through an interactive dashboard. Unlike the earlier exploratory phase, the Auckland implementation emphasized operational forecasting, real-time prediction capability, and deployment-oriented system design.

5.2 Auckland Data Processing Pipeline

The Auckland implementation used two primary processed datasets: The **auckland_final_ready** dataset represented the larger processed historical parking dataset used primarily for generalized model training and occupancy pattern learning. The smaller **akl_final** dataset was later used for localized validation and evaluation.

The overall Auckland data processing workflow is shown below:

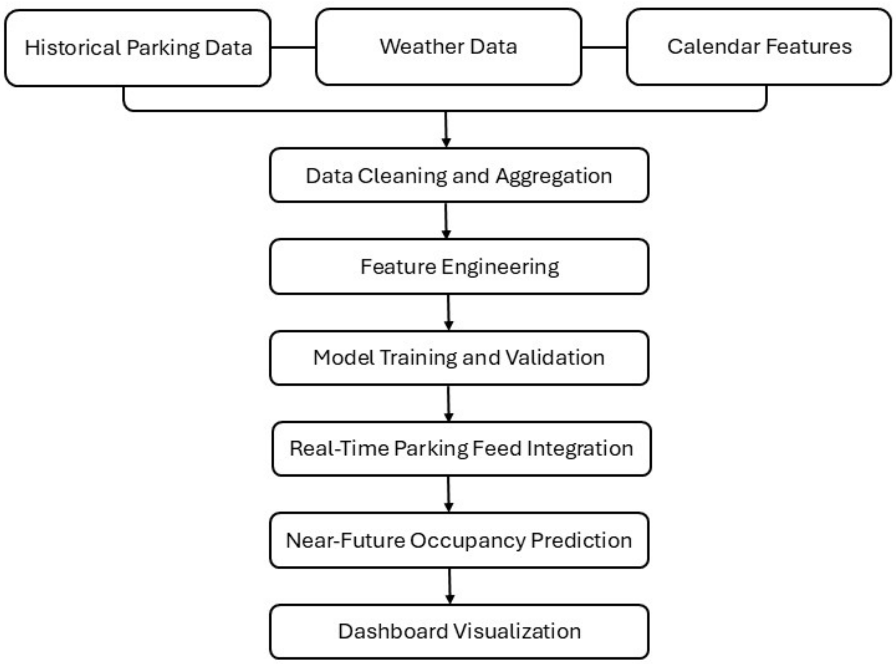

Figure 1: Data processing workflow

The processing pipeline involved timestamp standardization, occupancy normalization, hourly aggregation, feature engineering, and validation preparation.

Parking occupancy values were converted into normalized occupancy ratios ranging between 0 and 1. This allowed the system to maintain consistency across parking locations with different parking capacities.

Several pre-processing checks were also implemented to remove incomplete records, handle missing occupancy values, validate timestamp consistency, and ensure proper chronological ordering for time-series forecasting.

5.3 Feature Engineering Strategy

Feature engineering became one of the most important components of the Auckland forecasting framework. Earlier exploratory experiments showed that strong feature engineering significantly improved forecasting performance, sometimes more effectively than increasing model complexity alone.

The final Auckland system incorporated four major categories of predictive features: temporal features, weather-related features, occupancy history features, and interaction features.

5.4 Validation and Forecasting Design

One of the most important methodological improvements introduced during the Auckland phase involved the validation strategy.

Earlier exploratory experiments initially used random train-test splitting methods. However, this approach was later identified as less suitable for time-series forecasting because it could produce unrealistic validation behaviour. The final Auckland implementation used chronological time-based validation to better simulate realistic forecasting conditions.

The workflow followed chronological sorting → lag generation → rolling feature generation → removal of incomplete lag rows → time-based train-test splitting. This approach reduced the risk of temporal leakage and ensured that future occupancy information was not unintentionally introduced into the training process.

5.5 Real-Time Parking Prediction Framework

One of the major objectives of the Auckland implementation phase was integrating real-time parking information into the forecasting system. Real-time parking availability data was collected from Auckland Transport feeds for selected monitored facilities including Civic, Ronwood, Toka Puia, and Victoria Street.

The live parking feed provided current available parking counts, which were converted into occupancy ratios using known parking capacities. The operational workflow included collecting live occupancy data, generating temporal and weather features, reconstructing short-term occupancy context, and predicting next-hour parking occupancy levels.

The framework therefore supported near-future occupancy forecasting using both historical learned behaviour and current real-time parking conditions.

5.6 Forecasting for Non-Live Parking Locations

Live occupancy feeds were available only for selected Auckland parking facilities. To extend forecasting coverage, the framework also supported occupancy prediction for locations without real-time monitoring. For these locations, predictions were generated using historical occupancy behaviour, temporal patterns, weather conditions, and occupancy-history features derived from the larger Auckland historical dataset.

This approach allowed the forecasting framework to support both:

- real-time prediction for monitored parking facilities,
- and generalized forecasting for non-live parking locations.

5.7 Dashboard Visualization Design

To support operational interpretation and smart city visualization, an interactive Auckland parking dashboard was developed using Microsoft PowerBI platform.

The dashboard displayed parking locations across Auckland, occupancy levels, live monitored parking locations, and predicted near-future occupancy behaviour.

5.8 Key Outcomes of the Auckland Implementation

The Auckland implementation phase transformed the project from an exploratory machine learning study into a localized operational smart parking forecasting framework.

Several important outcomes were achieved during this phase:

- successful integration of historical and live parking data,
- development of a stable forecasting pipeline,
- implementation of time-based validation,
- integration of weather and temporal analytics,
- real-time occupancy prediction capability,
- and dashboard visualization.

The implementation phase also demonstrated that carefully engineered machine learning systems can achieve effective near-future parking occupancy forecasting even under practical urban constraints such as limited live sensor coverage and heterogeneous parking datasets.

Most importantly, the Auckland phase established a complete end-to-end smart parking prediction framework combining data engineering, predictive analytics, real-time integration, and operational visualization within a unified urban analytics system.

6. Modelling and Implementation

6.1 Overview

The implementation phase focused on transforming the processed Auckland datasets into a complete occupancy prediction pipeline capable of real-time occupancy monitoring and prediction for near future.

The implementation process included:

- target generation,
- temporal forecasting preparation,
- lag-based feature engineering,
- model benchmarking,
- validation strategy refinement,
- and real-time inference integration.

The final implementation prioritized forecasting stability, realistic validation behaviour, operational simplicity, and real-time prediction capability rather than focusing only on maximizing model complexity.

6.2 Forecasting Formulation

The parking occupancy problem was formulated as a supervised regression task. The objective was to predict the occupancy ratio for the next hour using historical occupancy behaviour, temporal features, and contextual environmental information.

The target variable was generated using a one-step future shift approach:

target_next = occupancy_rate.shift(-1). This allowed the model to learn relationships between current parking conditions and future occupancy behaviour.

The forecasting formulation evolved significantly during implementation. Earlier experiments focused mainly on predicting current occupancy behaviour. However, the system later transitioned toward true near-future forecasting through the introduction of future target generation, lag features, and rolling occupancy statistics.

6.3 Feature Preparation and Implementation

The final forecasting implementation used a combination of temporal, environmental, and occupancy-history features.

Temporal Features

The implemented temporal variables included hour, day_of_week, month, is_weekend, hour_sin, and hour_cos. Cyclical encoding using sine and cosine transformations improved the model's ability to capture periodic occupancy behaviour across different time periods.

Weather Features

Weather information was incorporated using temperature, precipitation, is_rain, and categorized rainfall levels. Additional interaction features were also implemented such as temp_hour, rain_hour, and hour_weekend.

These features attempted to model how environmental conditions influenced parking demand under different temporal conditions.

Lag-Based Features

One of the most important improvements introduced during implementation involved occupancy-history modeling. The implemented lag features included lag_1, lag_2, lag_3, and rolling_mean_3. The lag variables allowed the model to capture short-term occupancy continuity, while the rolling mean feature improved stability during fluctuating occupancy periods.

The final implementation workflow followed chronological sorting → future target creation → lag generation → rolling feature generation → removal of incomplete lag rows → train-test splitting. This ordering was important because it reduced the risk of temporal leakage during forecasting evaluation.

The primary forecasting framework used lag-based occupancy features and rolling occupancy statistics to capture short-term parking behaviour. However, for locations where sufficient occupancy history or live occupancy context was unavailable, a secondary no-lag forecasting configuration was also implemented using mainly temporal and weather-related features. This improved the operational flexibility of the forecasting system for locations with limited real-time occupancy information.

6.4 Validation Strategy

During the earlier exploratory phase, several experiments initially used random train-test splitting approaches. However, this method was later identified as less suitable for time-series forecasting because random splitting may produce unrealistic forecasting conditions.

The final implementation adopted chronological time-based validation. The validation workflow used earlier observations for training and later observations for testing. This approach better simulated real-world forecasting scenarios where future occupancy values are predicted using past information only.

6.5 LightGBM Implementation

LightGBM became the primary forecasting model used in the final Auckland implementation.

The model was selected because of:

- strong handling of structured tabular data,
- efficient training performance,
- ability to model non-linear relationships,
- and suitability for real-time inference.

The implemented LightGBM model used the engineered temporal, weather, and lag-based features to generate next-hour occupancy forecasts.

The model parameters were selected through iterative experimentation during the implementation phase rather than extensive hyperparameter optimization.

The final LightGBM configuration used:

- n_estimators = 300
- learning_rate = 0.05
- max_depth = 6

The relatively low learning rate improved forecasting stability by allowing the model to learn occupancy behaviour gradually across training iterations. Increasing the number of estimators to 300 helped compensate for the slower learning process and improved overall prediction consistency.

A moderate maximum tree depth of 6 was selected to reduce excessive model complexity and minimize overfitting risk under chronological validation conditions. This configuration also maintained efficient training and inference performance, which was important for supporting near-real-time occupancy prediction within the operational forecasting pipeline.

6.6 Benchmarking Experiments

Additional benchmarking experiments were conducted using Linear Regression, Random Forest, and LightGBM to evaluate forecasting accuracy, prediction stability, and operational suitability. The experiments showed that all three models achieved relatively competitive performance when combined with engineered temporal, weather, and occupancy-history features.

Linear Regression provided a strong baseline despite its simplicity, indicating that the forecasting features captured substantial parking occupancy behaviour patterns. Random Forest and LightGBM demonstrated greater flexibility for modelling non-linear occupancy relationships and feature interactions (Breiman, 2001; Ke et al., 2017).

Following iterative refinement of pre-processing, feature engineering, and chronological validation, LightGBM demonstrated the most stable overall forecasting behaviour across both operational and validation experiments.

7 Results and Evaluation

7.1 Final Forecasting Performance

The final Auckland forecasting framework was evaluated using a time-based train-test split to simulate realistic forecasting conditions. The model was trained using earlier observations and evaluated on later unseen occupancy records.

The final LightGBM implementation achieved the following results.

Final Forecasting Results

Metric	Value	Metric	Value
MAE	0.1298	RMSE	0.1893

Additional validation experiments conducted using a smaller Auckland dataset achieved MAE and RMSE values of 0.1214 and 0.1470 respectively, providing further evidence of the effectiveness of the forecasting framework under controlled evaluation conditions.

7.2 Benchmarking Evaluation

To evaluate the effectiveness of the forecasting framework, benchmarking experiments were conducted using Linear Regression, Random Forest, and LightGBM models. The evaluation focused on forecasting accuracy, error stability, and operational suitability for near-future parking occupancy prediction.

Model	MAE	RMSE

Linear Regression	0.1261	0.1798
Random Forest	0.1302	0.1781
LightGBM	0.1298	0.1893

The benchmarking results showed that all evaluated models achieved relatively competitive forecasting performance when combined with temporal, weather, and occupancy-history features. Linear Regression provided a strong baseline despite its simplicity, indicating that the engineered features captured a substantial proportion of parking occupancy behaviour.

Although Linear Regression achieved competitive benchmark accuracy, Random Forest and LightGBM provided greater flexibility for modelling non-linear occupancy relationships and heterogeneous feature interactions. LightGBM also offered better suitability for operational forecasting and future system expansion.

Overall, LightGBM demonstrated the most stable forecasting behaviour across both operational and validation experiments while providing greater flexibility. As a result, it was selected as the primary forecasting model for the final Auckland smart parking framework because of its scalability, flexibility, and suitability for real-time forecasting.

7.3 Location-Wise Forecasting Performance

Additional evaluation was conducted across selected Auckland parking locations to observe forecasting consistency under different parking conditions.

Location-Wise Forecasting Results

Location	**MAE**	**Location**	**MAE**
Civic	0.1347	Toka Puia	0.1392
Ronwood	0.1052	Victoria Street	0.1303

An example real-time prediction output is shown below.

Parking Location	**Occupancy Now**	**Predicted Next Hour**
Civic	0.9366	0.8625
Ronwood	0.2257	0.4757
Toka Puia	0.0714	0.3214
Victoria Street	0.3844	0.6344

The results showed moderate forecasting variation between locations. Parking facilities with relatively stable occupancy behaviour generally produced lower forecasting errors, while more dynamic parking environments generated slightly higher forecasting variability. Ronwood achieved the lowest forecasting error among the evaluated locations, while Toka Puia and Civic produced comparatively higher MAE values due to stronger occupancy fluctuations and more dynamic parking behaviour.

7.4 Actual vs Predicted Occupancy Analysis

To visually evaluate forecasting behaviour, actual occupancy values were compared against predicted next-hour occupancy values for selected parking locations.

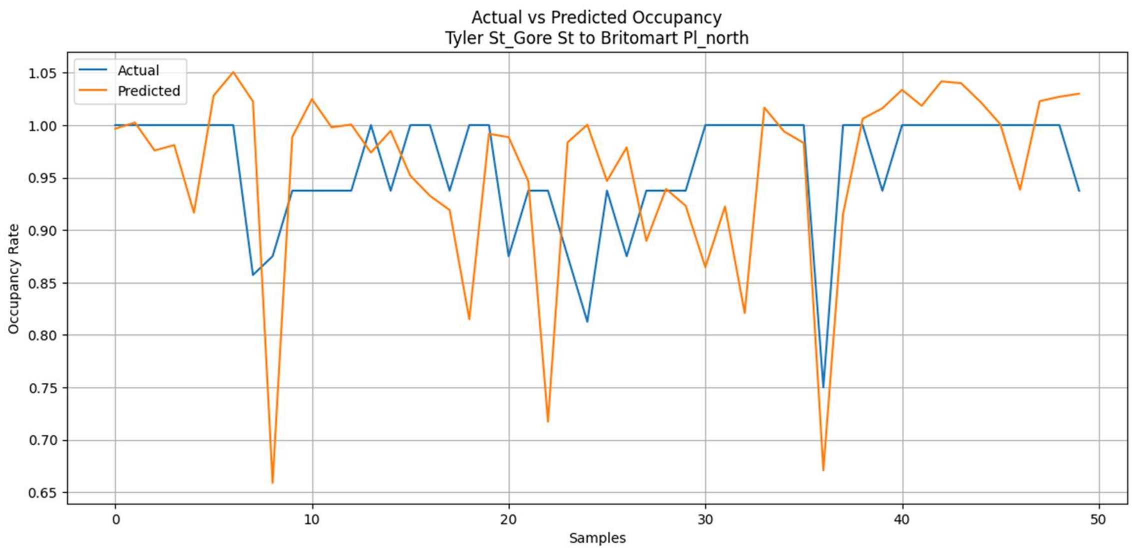

Figure 2: Actual vs Predicted Occupancy for a Selected Auckland Parking Location

The evaluation plots showed that the forecasting system successfully captured overall occupancy trends and short-term fluctuations while maintaining stable prediction behaviour. The predicted values generally remained close to the actual occupancy patterns, particularly during high-occupancy periods.

Some smoothing was observed during sudden occupancy changes. Overall, the results demonstrated that the framework could effectively model parking occupancy behaviour and generate stable near-future predictions.

7.5 Feature Importance Evaluation

Feature importance analysis was conducted using the trained LightGBM model to evaluate the contribution of individual predictive features.

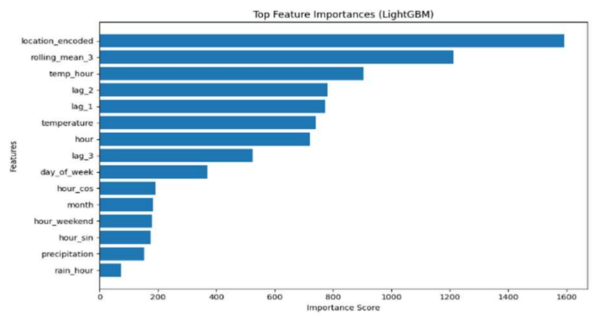

Figure 3: Top Feature Importance Generated by the LightGBM Forecasting Model

The most influential forecasting features included location_encoded, rolling_mean_3, lag_1, lag_2, temp_hour, temperature, and hour. The results showed that location-based information was the strongest predictor, confirming that parking occupancy behaviour varies significantly across different urban locations.

Occupancy-history features such as rolling_mean_3, lag_1, and lag_2 also contributed strongly, demonstrating that recent occupancy behaviour has a major influence on near-future parking demand. Temporal features including hour, hour_sin, hour_cos, and day_of_week further improved forecasting performance by capturing recurring parking patterns.

Weather-related variables such as temperature, precipitation, and rain_hour provided moderate but useful contextual information.

7.6 Residual Error Analysis

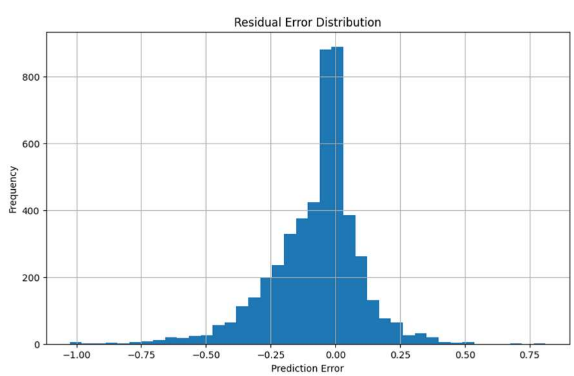

Figure 4: Residual Error Distribution of Forecasting Model

Figure 5: Residuals Versus Predicted Occupancy Values

These showed that most prediction errors were concentrated near zero, indicating relatively stable forecasting performance across the evaluation dataset. The residual histogram followed an approximately bell-shaped distribution, suggesting that the model did not exhibit severe systematic prediction bias and that most predictions remained within moderate error ranges.

The residual scatter plot further showed that residuals were generally centred around zero, while prediction variability increased slightly near full-capacity occupancy conditions. This behaviour is expected in parking forecasting because sudden occupancy changes and irregular urban activity are more difficult to predict accurately. Overall, the residual analysis demonstrated that the forecasting framework maintained stable operational prediction behaviour across most parking conditions.

7.7 Real-Time Prediction Evaluation

The final system was also evaluated using live Auckland Transport parking availability feeds.

Real-time occupancy predictions were generated for selected monitored parking facilities including Civic, Ronwood, Toka Puia, and Victoria Street.

Location	Occupancy	Predicted Next Hour
Civic	0.9366	0.8625
Ronwood	0.2257	0.4757
Toka Puia	0.0714	0.3214
Victoria Street	0.3844	0.6344

The real-time evaluation demonstrated that the forecasting framework could successfully integrate live parking availability data, historical occupancy learning, weather information, and temporal forecasting logic within a unified operational prediction pipeline.

8. Dashboard and Visualization

An interactive smart parking analytics dashboard was developed using Microsoft Power BI to support real-time parking monitoring, occupancy forecasting, spatial hotspot analysis, and smart city decision-making. The dashboard integrates historical parking occupancy data, live Auckland Transport parking feeds, weather information, temporal analytics, and machine learning forecasting outputs within a unified visualization framework.

8.1 Live Operations and Predictive Analytics Dashboard

The Live Operations Dashboard provides real-time parking monitoring and operational analytics for selected Auckland parking facilities. The dashboard integrates current occupancy levels, available parking spaces, next-hour occupancy forecasts, and live refresh capability.

Figure 6: Live Operations and Predictive Analytics Dashboard

- Real-time occupancy monitoring
- Machine learning occupancy forecasting
- Street-wise parking utilization analysis
- Hourly demand pattern visualization
- Spatial parking hotspot mapping
- AI-generated operational insights

8.2 Forecasting and Machine Learning Analytics Dashboard

The forecasting dashboard focuses on occupancy prediction trends, confidence analysis, forecast comparison, and machine learning model evaluation. The implemented LightGBM model uses temporal, weather, and occupancy-history features to generate near-future occupancy predictions.

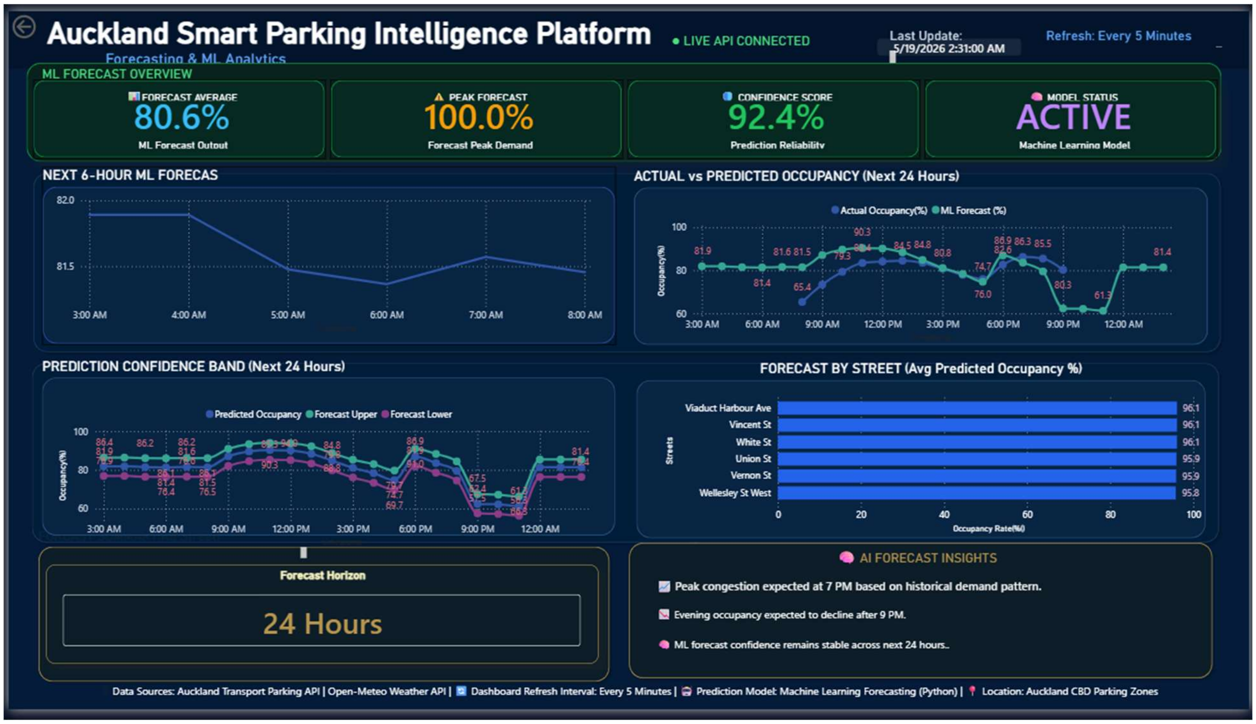

Figure 7: Forecasting and Machine Learning Analytics Dashboard

8.3 Spatial Parking Intelligence Dashboard

The spatial intelligence dashboard visualizes parking demand distribution and congestion hotspots across Auckland using GIS-based mapping and occupancy categorization.

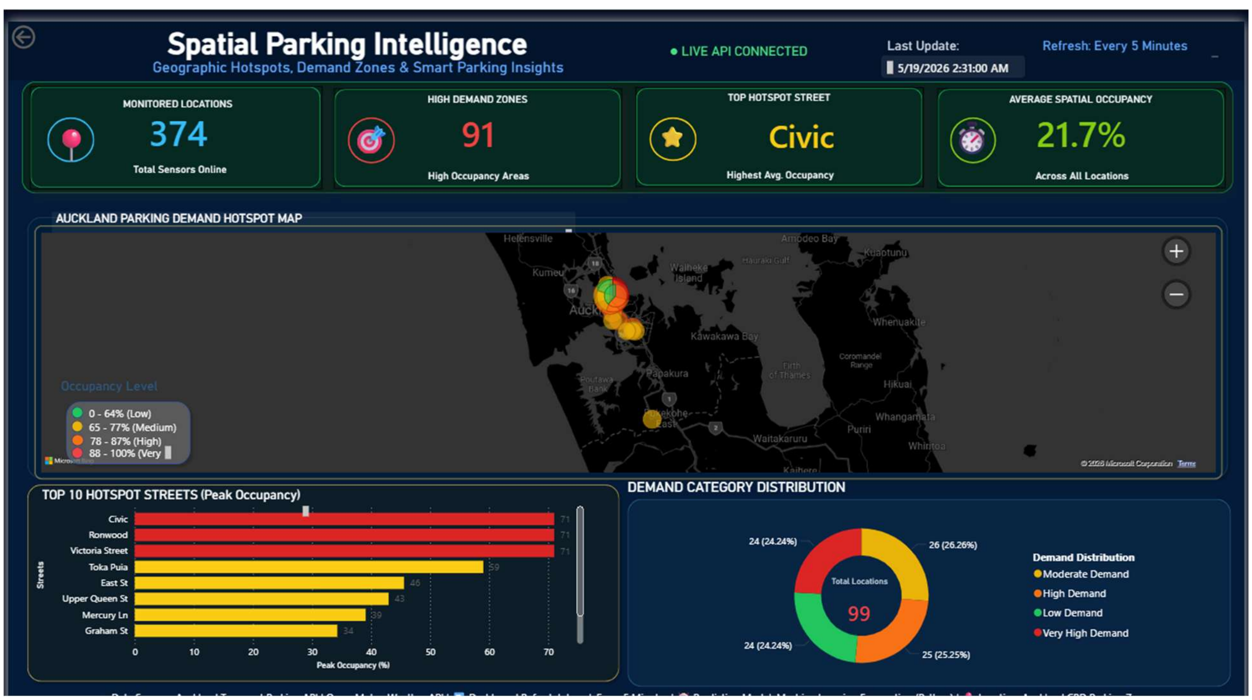

Figure 8: Spatial Parking Intelligence Dashboard

8.4 Dashboard Development Workflow

- Export of processed machine learning prediction datasets from Google Colab into structured CSV format
- Import and modelling of prediction datasets within Microsoft Power BI
- Development of KPI monitoring cards for occupancy and hotspot analysis
- Hourly demand pattern visualization
- Creation of interactive forecasting and parking utilization visualizations
- Integration of geographic latitude and longitude data for GIS-based spatial mapping
- Development of parking demand categorization and hotspot identification logic
- Implementation of color-coded occupancy and congestion visualization techniques
- Customization of dashboard layout, smart city theme, and interactive user interface components
- Validation and testing of dashboard visuals, filters, forecasting outputs, and spatial analysis results
- Final deployment of interactive smart parking intelligence dashboards for analytical reporting and decision support

8.5 Operational Value of the Dashboard

The dashboard demonstrates how machine learning, real-time urban analytics, and interactive visualization can support operational smart parking management. The framework provides parking forecasting, congestion monitoring, spatial demand analysis, and live operational visibility for urban mobility management.

9. Discussion

The project demonstrated that machine learning forecasting combined with temporal analytics, occupancy-history modelling, weather integration, and real-time parking feeds can support operational smart parking analytics and near-future occupancy prediction.

9.1 Forecasting Performance and Feature Influence

The evaluation results showed that temporal occupancy behaviour and occupancy-history variables were the strongest forecasting contributors. Features such as lag variables, rolling occupancy statistics, hour-of-day patterns, and weekday behaviour significantly improved prediction accuracy.

Although LightGBM achieved the best overall forecasting performance, simpler models also produced competitive results when combined with effective feature engineering and chronological validation strategies.

9.2 Localization and Cross-City Transferability

One of the most important findings of the project emerged during the exploratory cross-city experiments. These experiments demonstrated that parking behaviour varies significantly between urban environments. Models trained using one city's data could not reliably generalize to another city without additional adaptation and calibration.

The Auckland calibration experiments confirmed that localized forecasting strategies substantially improved prediction accuracy and operational forecasting stability. .

9.3 Real-Time Integration and System Limitations

The final forecasting framework successfully integrated machine learning prediction with real-time Auckland parking feeds and Power BI dashboard visualization. .

Several practical challenges were identified during implementation, including:

- heterogeneous datasets,
- inconsistent location naming,
- limited sensor coverage,
- and real-time feature consistency.

The system mainly focused on next-hour occupancy forecasting and did not fully capture irregular events or sudden occupancy fluctuations.

9.4 Reflection on the Research Process

The research evolved from exploratory Australian parking experiments toward a localized Auckland smart parking forecasting framework. Iterative experimentation, feature engineering refinement, cross-city validation, and operational dashboard development gradually shaped the final system implementation.

10. Conclusion and Future Work

10.1 Conclusion

This project investigated the development of a machine learning-based smart parking forecasting framework capable of predicting near-future parking occupancy using historical parking data, temporal analytics, weather information, and real-time parking feeds.

The research initially began as an exploratory cross-city study using parking datasets from Geelong, Melbourne, and Brisbane. This phase focused on occupancy behaviour analysis, feature engineering, cross-city validation, and calibration experiments. The results showed that parking occupancy behaviour differs significantly across cities, highlighting the importance of localized forecasting approaches.

The later availability of Auckland Transport historical and live parking datasets enabled the project to evolve into a more operational smart parking forecasting system focused on Auckland. The final framework integrated:

- machine learning forecasting,
- temporal and weather features,
- occupancy-history modelling,
- real-time parking feeds,
- and dashboard visualization.

Several machine learning models including Linear Regression, Random Forest, and LightGBM were evaluated. The final LightGBM framework achieved stable forecasting performance, with MAE values ranging from 0.1214 to 0.1298 and RMSE values ranging from 0.1470 to 0.1893 across the evaluation experiments.

The project also demonstrated successful real-time parking prediction and interactive dashboard visualization for selected Auckland parking locations.

10.2 Future Work

Although the project achieved stable forecasting performance and successful real-time integration, several opportunities remain for future improvement. The current implementation focused mainly on next-hour occupancy forecasting, while future work could explore:

- multi-hour forecasting,
- long-term parking demand prediction,
- and more advanced temporal forecasting approaches.

Future improvements could also include expanding real-time integration across more Auckland parking locations and incorporating additional urban factors such as traffic congestion, public events, road incidents, and public transport activity.

The project also creates opportunities to investigate advanced deep learning methods including:

- LSTM networks,
- Transformer architectures,
- and hybrid forecasting models.

In addition, future systems could be developed as fully operational web-based smart parking platforms with live prediction updates, parking recommendations, and route optimization features. Further research on transfer learning and cross-city adaptation could also improve forecasting portability across different urban environments. Overall, the project provides a strong foundation for future smart parking analytics and operational smart city forecasting systems.

11. Personal Learning Reflection

Shovan Barai

This project significantly enhanced my understanding of machine learning methodologies, feature engineering techniques, API integration processes, dashboard development, and smart city analytics. I gained valuable practical experience in integrating Python-based analytical workflows with Power BI visualisation platforms to develop intelligent transportation system applications. The project also strengthened my problem-solving, data integration, and predictive modelling skills within real-world transportation analytics contexts.

Amit Chakrabarty

This project improved my understanding of collaborative system development, transportation analytics, and dashboard-based decision support systems. I gained practical experience in integrating multiple technologies, analysing transportation datasets, and contributing to the development of predictive analytics solutions for smart parking management. The project also enhanced my knowledge of intelligent transportation systems and data-driven urban mobility analysis.

S M Amit Hasan

This project enhanced my understanding of predictive analytics, machine learning workflows, and intelligent transportation systems. Through participation in the project, I gained practical experience in data pre-processing, feature engineering, machine learning model implementation, and analytical visualisation techniques. The project further improved my technical understanding of how predictive technologies can support smart city and transportation management applications.

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13. Appendix

13.1 Email Response from Auckland Transport

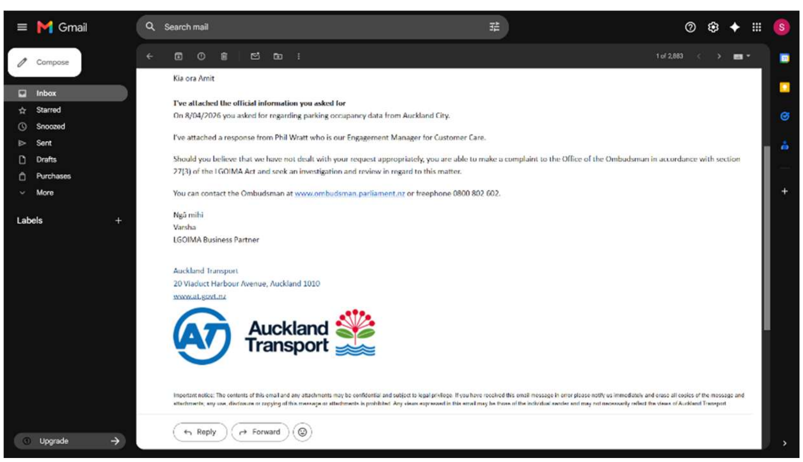

Figure 9: Email Response from Auckland Transport

13.2. GitHub Repository:

https://github.com/shovanbarai/Auckland-Smart-Parking-Intelligence

13.3 Final Model Parameters

```python
# ---- MODEL ----
model = LGBMRegressor(
    n_estimators=300,
    learning_rate=0.05,
    max_depth=6
)
```

Figure 10: Final Model Parameters

13.4 Prediction Output Sample for Live Locations

```
==============================
 FINAL REAL-TIME PREDICTION
==============================
        location_id  occupancy_now  predicted_next_hour  predicted_cars
1             Civic       0.870968             0.746931           695.0
3           Ronwood       0.221239             0.421239           286.0
4         Toka Puia       0.059524             0.259524           109.0
5   Victoria Street       0.557542             0.746931           669.0
```

Figure 11: Prediction Output Sample for Live Locations

13.5 Prediction Output Sample for Non-Live Locations

```
==============================
 OTHER LOCATIONS (FINAL FORMAT)
==============================
                                        location_id  occupancy_now  \
0              Adelaide St_Drake St to Sale St_east       0.666667
1             Adelaide St_Drake St to Sale St_south       0.666667
2        Airedale St_Lyndock St to St Paul St_north       1.000000
3        Airedale St_Lyndock St to St Paul St_south       1.000000
4    Airedale St_Queen St to end of Airdale St_north      0.800000
5    Airedale St_Queen St to end of Airdale St_south      0.000000
6   Airedale St_Queen Street to end (dead end)_north      0.333333
7   Airedale St_Queen Street to end (dead end)_south      0.000000
8         Airedale St_Symonds St to Lyndock St_east       1.000000
9     Airedale St_Symonds St to Lyndock St_northeast      0.000000

   predicted_next_hour  predicted_cars
0             0.219812            15.0
1             0.720377            19.0
2             0.720377            69.0
3             0.698144            67.0
4             0.760464            61.0
5             0.330092            26.0
6             0.508676            26.0
7             0.508676            26.0
8             0.664303            24.0
9             0.664303            31.0
```

Figure12: Prediction Output Sample for Non-Live Locations